Oscillatory filling sums with independently long twists

Abstract

An exact finite Hermite functional can be written as a sum of oscillatory integrals indexed by nonzero integer Fourier charges. We prove an asymptotic formula for the whole sum from two source estimates: a nonvanishing holomorphic expansion near the complete point and a strict spectral gap away from it. The twist numbers are independently signed, sufficiently large integers, fixed before the level tends to infinity. The proof first bounds all distant charges on the original intervals, including the paired endpoint contributions. Only the remaining finite set is localized near its stationary points. The charges of magnitude one contribute equally, while every other charge is exponentially smaller. The resulting amplitude is nonzero and retains the finite-level action. We state the application to marked knot sources with its local analytic and geometric inputs explicit; this paper does not prove those inputs or estimate the other degree regions of the invariant.

1 The question and the hypotheses

The volume conjecture concerns an oscillatory invariant, so an estimate for a positive sum of absolute values does not determine its limit. Even after a promising geometric contribution has been isolated, one must show that it occurs on the original contour and survives the other terms of the same sum. We address one precise part of that problem: the passage from a local holomorphic source expansion to its entire Fourier filling sum.

The motivating finite functional is defined in [1]. It includes first derivatives at double spectral roots, as well as the value at an odd simple root. Its Fourier representation involves every nonzero integer charge. Absolute convergence for each fixed level is insufficient for an asymptotic argument: the constants in such a convergence estimate can grow exponentially with the level. We therefore prove a separate estimate for the infinite set of distant charges before restricting any integral to a small neighborhood.

Our contour and Gaussian arguments are instances of the classical saddle-point method; see, for example, [2, §2.4] and [3]. The proof below gives the contour, its side faces, the determinant branch and the charge estimates explicitly. These details are needed to distinguish the complete filling sum from an arbitrarily chosen local integral.

Fix an integer $r \geq 1$. For every integer $N \geq 3$, let $P_N(\sigma_1, \dots, \sigma_r)$ be a complex polynomial of degree at most $N-1$ in each separate variable. Put

$$C_N(\delta) = P_N(-2\cos(2\pi\delta_1), \dots, -2\cos(2\pi\delta_r)), \qquad \delta = (\delta_1, \dots, \delta_r) \in \mathbb{C}^r. \tag{1.1}$$

Thus C_N is entire, one-periodic and even in each variable separately. No bound on its total degree smaller than $r(N-1)$ is assumed. Write $I = [-1/2, 1/2]$, and use Euclidean norm for real vectors and maximum norm when specifying polydisks.

We make the following two analytic assumptions. Their constants do not depend on the later twist numbers.

(i) There is a fixed complex polydisk about zero, a real exponent α, unit scalars ε_N, and holomorphic functions A_N, a_N, e_N on that polydisk such that

$$C_N(\delta) = \varepsilon_N N^{\alpha} a_N(\delta) e^{NA_N(\delta)}(1 + e_N(\delta)), \qquad \sup|e_N| = O(N^{-1}). \tag{1.2}$$

The functions A_N are separately even and real on the real polydisk. They converge locally uniformly to a holomorphic function A_0, and a_N converge locally uniformly to a holomorphic function a with $a(0) \neq 0$. Finally, for real numbers Φ and $c_j > 0$,

$$A_0(0) = \Phi, \qquad A_0''(0) = -2\,\mathrm{diag}(c_1, \dots, c_r). \tag{1.3}$$

All functions are considered on smaller closed polydisks when a uniform bound is used. Cauchy's formula then gives convergence and bounds for every fixed derivative. Shrinking the disk makes a and, for all large N, a_N nonzero there.

(ii) For each sufficiently small $\eta > 0$ there are numbers $\tau_\eta, \gamma_\eta > 0$ such that, for all large N,

$$|C_N(x + \mathrm{i}y)| \le e^{N(\Phi - \gamma_\eta)} \quad \text{when } x \in I^r, \quad \max_j |x_j| \ge \eta, \quad \max_j |y_j| \le \tau_\eta. \tag{1.4}$$

The bound is for the shared polynomial P_N itself. Separate estimates for its local factors are not a substitute for this hypothesis.

The restriction to sufficiently small η loses nothing in later uses at an endpoint: an endpoint also lies outside any such small neighborhood. Periodicity permits equivalent real representatives. Bounds for fixed derivatives follow on smaller strips by Cauchy's formula, keeping one separated real coordinate at distance at least $\eta/2$ from the origin modulo the period.

For a fixed tuple $\boldsymbol{n} = (n_1, \dots, n_r)$ of nonzero signed integers, set $d = N - 1$, $q = e^{\pi \mathrm{i}/N}$ and

$$\Gamma_N(\boldsymbol{n}) = \exp\left(\frac{\pi \mathrm{i} d^2}{4N} \sum_j n_j\right). \tag{1.5}$$

It has modulus one; the exponential specifies its lift without a choice of fractional power. Define

$$\begin{aligned} \mathcal{I}_N(\boldsymbol{n}) = \Gamma_N(\boldsymbol{n})(-\mathrm{i})^r \sum_{\ell \in (\mathbb{Z}\setminus\{0\})^r} \left(\prod_j \ell_j\right) (-1)^{(N+1)\sum_j \ell_j} \\ \times \int_{I^r} C_N(\delta) \prod_j \sin(2\pi\delta_j) \exp\left\{N \sum_j (\pi \mathrm{i} n_j \delta_j^2 + 2\pi \mathrm{i} \ell_j \delta_j)\right\} d\delta. \end{aligned} \tag{1.6}$$

Section 2 proves absolute convergence at each fixed N and identifies this expression with the finite Hermite functional. The term *charge* will mean precisely an index ℓ_j in this sum.

Theorem 1.1 (The whole filling sum)**.** *Under (1.1)–(1.4), there is n_0 such that the following holds for every fixed signed integer tuple with $|n_j| \ge n_0$ for all j. For all sufficiently large N, the phase*

$$F_{N,\ell}(\delta) = A_N(\delta) + \pi \mathrm{i} \sum_j n_j \delta_j^2 + 2\pi \mathrm{i} \sum_j \ell_j \delta_j \tag{1.7}$$

has a unique critical point near zero for $\ell = \mathbf{1} = (1, \dots, 1)$. Write that point as z_N, and put

$$S_N = F_{N,\mathbf{1}}(z_N), \qquad Q_N = -F''_{N,\mathbf{1}}(z_N).$$

Then

$$\begin{aligned} \mathcal{I}_N(\boldsymbol{n}) = \Gamma_N \varepsilon_N (-1)^{(N+1)r} (-2\mathrm{i})^r (2\pi)^{r/2} N^{\alpha - r/2} \\ \times \frac{a_N(z_N) \prod_j \sin(2\pi z_{N,j})}{\sqrt{\det Q_N}} e^{N S_N} \left(1 + O_{\boldsymbol{n}}(N^{-1}) + O_{\boldsymbol{n}}(e^{-\kappa_{\boldsymbol{n}} N})\right) \end{aligned} \tag{1.8}$$

for some $\kappa_{\boldsymbol{n}} > 0$. The square root is the Gaussian branch on symmetric matrices with positive definite real part. The critical points, actions and normalized coefficients converge; the coefficient in the second line, excluding e^{NS_N}, converges to a nonzero number. In particular $\mathcal{I}_N$ is eventually nonzero and

$$\lim_{N\to\infty} \frac{1}{N} \log |\mathcal{I}_N(\boldsymbol{n})| = \operatorname{Re} S_0, \qquad S_0 = \lim_{N\to\infty} S_N. \tag{1.9}$$

Moreover $\Phi - \operatorname{Re} S_0 = O(\sum_j n_j^{-2})$ as the independent magnitudes tend to infinity. Neither the threshold nor this last bound imposes a restriction on their ratios.

The constants in the relative error may depend on the fixed tuple. The theorem makes no assertion when the twists grow with N. The finite S_N remains inside the exponential. Convergence $S_N \to S_0$ alone would not justify replacing e^{NS_N} by e^{NS_0} in a relative asymptotic formula.

1.1 An elementary example

The two assumptions can already be checked for the polynomial sequence

$$P_N(\sigma) = \prod_{j=1}^{r} (\beta_j - \gamma_j \sigma_j/2)^{N-1}, \qquad \beta_j > \gamma_j > 0.$$

Here

$$C_N(\delta) = \prod_j (\beta_j + \gamma_j \cos(2\pi\delta_j))^{N-1}.$$

Choose the logarithm that is real at zero. On a fixed sufficiently small complex polydisk the factors are nonzero, and assumption (i) holds exactly with $\alpha = 0$, $\varepsilon_N = 1$, $a_N = 1$, $e_N = 0$ and

$$A_N(\delta) = (1 - N^{-1}) \sum_j \log(\beta_j + \gamma_j \cos(2\pi\delta_j)).$$

In particular,

$$\Phi = \sum_j \log(\beta_j + \gamma_j), \qquad c_j = \frac{2\pi^2 \gamma_j}{\beta_j + \gamma_j} > 0.$$

On the real torus each factor is positive and attains its maximum only at $\delta_j = 0$ modulo one. Compactness gives a number $g > 0$ for which the product is at most $e^{\Phi - g}$ outside any fixed neighborhood of zero. Continuity bounds its modulus by $e^{\Phi - g/2}$ in a sufficiently thin complex strip. For all large N, its $(N-1)$st power is at most $e^{N(\Phi - g/4)}$. This proves assumption (ii). The theorem therefore applies to the entire filling sum of this explicit sequence. This example is not asserted to be a knot invariant. It illustrates why the local expansion and the global gap are distinct hypotheses and how each can be verified.

2 The exact Fourier representation

We explain the normalization before estimating the sum. For a polynomial H of degree at most d, the finite filling functional of [1, Section 3] has the midpoint formula

$$\mathcal{L}_N^n H = -N^{-2} \sum_{k=0}^{N-1} g'(\phi_k), \qquad \phi_k = \frac{(2k+1)\pi}{N}, \tag{2.1}$$

where

$$g(\phi) = \sin\phi\, q^{nd^2/4} \exp\left(\frac{inN(\phi-\pi)^2}{4\pi}\right) H(2\cos\phi), \qquad 0 \le \phi \le 2\pi.$$

The functional in (2.1) retains the derivative at each double root; it is not evaluation only at the distinct roots. Its finite proof is the only quantum-algebra input to this section.

At the endpoints, $g = 0$ and $g'(0) = g'(2\pi)$. The equality follows because the derivative of the sine is the only surviving term and the remaining factors have the same endpoint values. Let

$$\widehat{g}_k = \frac{1}{2\pi}\int_0^{2\pi} g(\phi) e^{-\mathrm{i}k\phi}\, d\phi.$$

Three integrations by parts, retaining the boundary difference in g'', give $\widehat{g}_k = O_{N,n}(|k|^{-3})$. Consequently the Fourier series of g' is absolutely convergent, with coefficients $\mathrm{i}k\widehat{g}_k$, and represents its continuous periodic extension. One way to verify the last assertion without differentiating a conditionally convergent series is to integrate by parts once to compute the Fourier coefficients of g'. The absolutely convergent series and g' then have the same Fourier coefficients; uniqueness follows, for example, by convolution with the finite Fejér kernels.

The finite geometric sum at the midpoints is zero unless $k = N\ell$, in which case it is $N(-1)^\ell$. Substitution in (2.1) therefore gives

$$\mathcal{L}_N^n H = -\mathrm{i}\sum_{\ell\neq 0} \ell(-1)^\ell \widehat{g}_{N\ell}.$$

Set $\phi = \pi - 2\pi\delta$. The Jacobian cancels the factor 2π in the Fourier coefficient, while $e^{-\pi\mathrm{i}N\ell} = (-1)^{N\ell}$. We obtain exactly

$$\begin{aligned}\mathcal{L}_N^n H = q^{nd^2/4}(-\mathrm{i})\sum_{\ell\neq 0} \ell(-1)^{(N+1)\ell} \\ \times \int_I \sin(2\pi\delta) H(-2\cos(2\pi\delta)) e^{N(\pi\mathrm{i}n\delta^2 + 2\pi\mathrm{i}\ell\delta)}\, d\delta.\end{aligned} \tag{2.2}$$

Applying this identity in each separate polynomial variable gives (1.6). The integrations-by-parts argument below proves absolute convergence of the multiple series as well. Thus the tensor functional and the multiple Fourier sum agree without an interchange of an uncontrolled limiting series.

Proposition 2.1 (Finite equality)**.** *For every P_N of the stated separate degrees,*

$$\mathcal{I}_N(\boldsymbol{n}) = \left(\bigotimes_{j=1}^{r} \mathcal{L}_N^{n_j}\right) P_N.$$

This statement is exact at each integer level $N \ge 3$ and for every signed integer tuple. The asymptotic hypotheses are unnecessary here.

3 Estimates on the original intervals

Lemma 3.1 (Polynomial growth off the real torus)**.** *There are real numbers $b_N \to 0$ such that*

$$|C_N(x+\mathrm{i}y)| \le \exp\left(N(\Phi + b_N) + 2\pi(N-1)\sum_j |y_j|\right) \quad (x, y \in \mathbb{R}^r). \tag{3.1}$$

Proof. On a small real cube the Hessian in (1.3), convergence and separate evenness imply $A_N(x) \le A_N(0) - c|x|^2$ for all large N. The amplitude and level power in (1.2) are subexponential. Outside a smaller cube use (1.4). This gives $\sup_{I^r} |C_N| \le e^{N(\Phi+b_N)}$ with $b_N \to 0$.

For one variable write $w = e^{2\pi \mathrm{i}\delta}$. The function $w^d H(-w - w^{-1})$ is a polynomial of degree at most $2d$. On the unit circle its modulus is at most $\|H\|_{[-2,2]}$. The exterior maximum principle gives a bound $|w|^{2d}\|H\|$ for $|w| \ge 1$. After division by w^d this is $|w|^d\|H\|$. The symmetry under $w \mapsto w^{-1}$ gives the corresponding bound for $|w| \le 1$. Apply this argument successively in the r variables to prove (3.1). □

Fixed source derivatives can be bounded by Cauchy's formula on a small fixed complex neighborhood. In contrast, derivatives of the Gaussian factor will be taken explicitly on real spectator variables. This distinction prevents large twist numbers from entering a complex-source bound where they do not belong.

Lemma 3.2 (Endpoint identities and summable charge weights)**.** *Let*

$$B_N(\delta) = C_N(\delta) \prod_j \sin(2\pi\delta_j) e^{\pi \mathrm{i} N \sum_j n_j \delta_j^2}.$$

For each coordinate, with all other variables fixed,

$$B_N|_{\delta_j = \pm 1/2} = 0, \qquad \partial_j B_N|_{\delta_j = 1/2} = \partial_j B_N|_{\delta_j = -1/2}. \tag{3.2}$$

These identities remain valid after arbitrary differentiations in other coordinates. Three integrations by parts give

$$\int_I B_N e^{2\pi \mathrm{i} N \ell_j \delta_j}\, d\delta_j = \frac{1}{(2\pi \mathrm{i} N \ell_j)^3} \left\{ [\partial_j^2 B_N e^{2\pi \mathrm{i} N \ell_j \delta_j}]_{-1/2}^{1/2} - \int_I \partial_j^3 B_N e^{2\pi \mathrm{i} N \ell_j \delta_j}\, d\delta_j \right\}. \tag{3.3}$$

In particular the sum (1.6) *is absolutely convergent at each fixed N and fixed $\boldsymbol{n}$.*

Proof. The first identity follows from the sine. In its first derivative only the cosine term survives; the source and the Gaussian have equal values at the two endpoints. This remains true with complex values of the other coordinates. The Fourier characters also have equal endpoint values because $N\ell_j$ is an integer. The first two boundary terms therefore cancel, giving (3.3). Repeating it in all coordinates gives a constant depending on $N, \boldsymbol{n}$ times $\prod_j |\ell_j|^{-3}$. After multiplication by $\prod_j |\ell_j|$, the majorant is a product of convergent $\sum_{\ell_j \ne 0} |\ell_j|^{-2}$ series. □

When (3.3) is applied only in a subset of coordinates, call them *spectator coordinates*. Each contributes at most three boundary or integral terms. On real spectators a Gaussian derivative of order at most three costs $O_{\boldsymbol{n}}(N^3)$. Cauchy bounds for the source derivatives, at a fixed radius, have the same exponential rate as the source bound being used, after any stated reduction of its gap. The N^3 cancels the denominator in (3.3); each weighted spectator charge thus contributes the summable factor $|\ell_j|^{-2}$. There are at most 3^{r-1} terms when all spectators of one distinguished coordinate are treated.

4 Critical points for independently signed twists

The following construction uses positive real charges. Negative charges will be restored by exact reflection symmetry, rather than by choosing new determinant branches.

Lemma 4.1 (The finite critical point)**.** *There are fixed numbers $B, a > 0$, with a sufficiently small relative to B, and a threshold n_0 such that, for positive real charges with $0 < \ell_j/|n_j| \le a$, the phase (1.7) has a unique critical point in a fixed neighborhood of zero. It satisfies*

$$z_j = -\frac{\ell_j/n_j}{1 + \mathrm{i}B_{N,j}(z)/(\pi n_j)}, \qquad \partial_j A_N(z) = -2z_j B_{N,j}(z), \tag{4.1}$$

and

$$|z_j| \le 2\ell_j/|n_j|, \qquad \operatorname{Im} z_j = \frac{\ell_j}{\pi n_j^2} \frac{\operatorname{Re} B_{N,j}(z)}{|1 + \mathrm{i}B_{N,j}(z)/(\pi n_j)|^2} > 0. \tag{4.2}$$

The matrix $Q_{N,\ell} = -F''_{N,\ell}(z)$ has positive definite real part, uniformly bounded below for all these charges and large N.

Proof. Separate evenness makes $\partial_j A_N$ divisible by z_j in its convergent power series, giving the holomorphic $B_{N,j}$. After a fixed shrink of the disk and an increase of the starting level, $|B_{N,j} - c_j| \le c_j/8$. Their first derivatives are uniformly bounded on a smaller disk. The right side of (4.1) maps the polydisk $|z_j| \le 2a$ into itself when n_0 is large. Its derivative norm is at most Ca/n_0, so the contraction theorem gives a unique fixed point there. Any critical point in the original small disk satisfies the displayed equation and hence lies in this polydisk. This proves the stated uniqueness.

Taking the imaginary part of that exact equation gives (4.2); it is positive for either sign of n_j. All denominators stay uniformly separated from zero. On a sufficiently small fixed complex disk, convergence and (1.3) give $-\operatorname{Re} A''_N \ge cI$. The twist Hessian is purely imaginary on real tangent vectors, so $\operatorname{Re} Q_{N,\ell} \ge cI$ as well. □

For each fixed twist and charge tuple, the contraction maps converge uniformly with their derivatives. Their fixed points therefore converge to those for A_0, as do the corresponding critical values and Hessians. No quantitative rate of this convergence is presumed.

Lemma 4.2 (Comparison of charge magnitudes)**.** *After enlarging n_0, for integer positive charges in the preceding box one has*

$$\operatorname{Re} S_{N,\mathbf{1}} - \operatorname{Re} S_{N,\ell} \ge \sum_j \frac{c_j}{2n_j^2}(\ell_j^2 - 1). \tag{4.3}$$

Also $|\operatorname{Re} S_{N,\mathbf{1}} - A_N(0)| \le C \sum_j n_j^{-2}$.

Proof. Allow the charges to vary continuously. Stationarity removes the derivatives of the critical point, so

$$\frac{\partial}{\partial \ell_j} \operatorname{Re} S_{N,\ell} = -2\pi \operatorname{Im} z_j.$$

Equation (4.2) and the fixed closeness to c_j give $\operatorname{Im} z_j \ge c_j \ell_j/(2\pi n_j^2)$, uniformly in all other charges in the box. Integrate successively from 1 to ℓ_j; every intermediate vector stays in the box. This proves (4.3) without comparing different twist sizes. For the last assertion allow all charges, including zero, in the same contraction. The upper estimate $0 \le \operatorname{Im} z_j \le C\ell_j/n_j^2$ and integration from the zero vector to $\mathbf{1}$ give the required bound. At zero charge the critical point is zero and the action is $A_N(0)$. □

5 The original local contour and its Gaussian coefficient

Let $K_B = [-B, B]^r$, with B fixed inside the local expansion disk. Choose a in Lemma 4.1 sufficiently small relative to B and r. For a local positive charge write $z = x_* + \mathrm{i}y_*$. Translate

K_B vertically to $K_B + \mathrm{i}y_*$, one coordinate at a time. The holomorphic integrand gives exactly the translated integral and its side-face integrals. The coefficient of the translated integral is one; it has not been supplied by a choice of a saddle contribution.

Lemma 5.1 (Side faces and the local expansion)**.** *Uniformly for the local charges of Lemma 4.1, the side-face integrals have exponential rate at most $\Phi - \gamma_B'$ for a fixed $\gamma_B' > 0$, up to polynomial factors and constants depending on the fixed twist tuple. For each such fixed charge,*

$$\begin{aligned}&\int_{K_B} C_N(\delta) \prod_j \sin(2\pi\delta_j) e^{N\sum_j (\pi \mathrm{i} n_j \delta_j^2 + 2\pi \mathrm{i} \ell_j \delta_j)} d\delta \\ &\quad = \varepsilon_N (2\pi)^{r/2} N^{\alpha - r/2} \frac{a_N(z) \prod_j \sin(2\pi z_j)}{\sqrt{\det Q_{N,\ell}}} e^{N S_{N,\ell}} (1 + O_{\boldsymbol{n},\ell}(N^{-1})) + O_{\boldsymbol{n},\ell}\big(N^C e^{N(\Phi - \gamma_B')}\big).\end{aligned} \tag{5.1}$$

The leading coefficient converges to a nonzero number for positive fixed charges.

Proof. Reality of A_N on the real chart and the vanishing of its first derivatives at zero imply, throughout the required small rectangles,

$$\operatorname{Re} A_N(x + \mathrm{i}y) \le A_N(0) - c|x|^2 + C|y|^2. \tag{5.2}$$

The term linear in y is purely imaginary. During a vertical translation, $0 \le y_j \le y_{*,j}$ and $|n_j| y_j \le C\ell_j / |n_j| \le Ca$ by (4.2). The real twist term is $-2\pi \sum_j n_j x_j y_j$, whose absolute value is at most $Ca \sum_j |x_j|$. The real charge term is $-2\pi \sum_j \ell_j y_j \le 0$. On a side face one real coordinate has modulus B. The loss cB^2 in (5.2) dominates the possible $O(raB) + O(ra^2/n_0^2)$ cost when a is sufficiently small. Since $A_N(0) \to \Phi$, this proves the fixed side-face gap. The local amplitudes and powers of N do not change it.

On the translated real cube, x_* is an interior point with all coordinates of modulus at most $2a$. The gradient of the entire complex phase vanishes there. Its real Hessian on real tangent vectors is uniformly negative definite. Thus

$$\operatorname{Re} F_{N,\ell}(x + \mathrm{i}y_*) \le \operatorname{Re} S_{N,\ell} - c'|x - x_*|^2.$$

The part separated from x_* is exponentially smaller. Put $t = \sqrt{N}(x - x_*)$ in a fixed small neighborhood. Taylor expansion gives the Gaussian $e^{-t^T Q_{N,\ell} t/2}$ and an error controlled by uniform derivatives of A_N and the holomorphic amplitude. The first cubic phase term and the first linear amplitude term are odd, so their integrals on a symmetric centered neighborhood vanish. The quartic term, the square of the cubic term and the remaining amplitude terms contribute $O(N^{-1})$. To justify termwise estimates, first restrict to $|t| \le N^\epsilon$, with any fixed sufficiently small $\epsilon > 0$, and use the Taylor remainder there; outside that region the preceding quadratic bound gives a superpolynomial tail. The $O(N^{-1})$ source error in (1.2), including its derivatives on a smaller disk, contributes the same relative order.

For a complex symmetric matrix Q with positive definite real part,

$$\int_{\mathbb{R}^r} e^{-t^T Q t/2}\, dt = (2\pi)^{r/2} (\det Q)^{-1/2}.$$

For real positive definite Q this follows from the ordinary real Gaussian change of variables. Both sides are holomorphic on the convex domain $\operatorname{Re} Q > 0$: the integral is locally dominated by a real Gaussian, and the nonzero determinant has a holomorphic logarithm fixed by its positive real values. Analytic continuation proves the formula on that domain. This specifies the square root used above.

Finally $a_N(z)$ has a nonzero limit, $Q_{N,\ell}$ tends to an invertible matrix with positive real part, and every z_j has a strictly positive imaginary part in the limit. Therefore none of the sine factors or determinants vanishes. This proves both the relative error assertion and nonvanishing of the limiting coefficient. □

The real integral outside K_B is bounded directly by (1.4); its chirp and charge factors have modulus one. This observation applies only to a fixed finite collection of charges. It is not used to split an infinite Fourier series at the boundary of K_B.

6 The infinite set of distant charges

We now estimate complete weighted charge terms in (1.6). Fix a small strip height $\tau > 0$ for which (1.4) gives a strict source bound at $\delta_i = \pm 1/2 + \mathrm{i}v$, $0 \le v \le \tau$, with the remaining variables real. Choose it smaller if necessary so that Cauchy neighborhoods for all required spectator derivatives retain a strict gap. Denote the resulting gap by $\gamma_e > 0$.

For a positive charge ℓ_i, shift its entire interval upward by $\mathrm{i}\tau$. For any entire integrand f, the oriented rectangle identity is

$$\int_I f(z)\,dz = \int_{I+\mathrm{i}\tau} f(z)\,dz - \mathrm{i}\int_0^\tau \bigl[f(1/2+\mathrm{i}v) - f(-1/2+\mathrm{i}v)\bigr]\,dv.$$

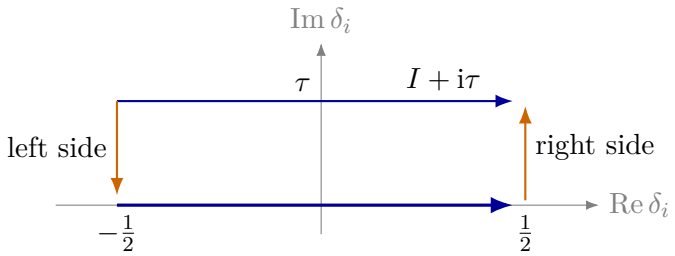
Both horizontal integrals run from left to right. The difference of the vertical sides is subtracted, as indicated by this identity. Let

$$B_i(z;\delta_{\neq i}) = \sin(2\pi z)e^{\pi \mathrm{i} n_i N z^2} C_N(z,\delta_{\neq i}).$$

Periodicity of C_N gives the exact difference of the two vertical endpoint integrands:

$$\begin{aligned} B_i(1/2+\mathrm{i}v) - B_i(-1/2+\mathrm{i}v) = {} & 2\mathrm{i}\sinh(2\pi v)\sinh(\pi n_i N v) \\ & \times e^{\pi \mathrm{i} n_i N(1/4-v^2)} C_N(1/2+\mathrm{i}v,\delta_{\neq i}). \end{aligned} \tag{6.1}$$

The other sine and Gaussian factors are left outside this identity. The endpoint Fourier characters agree, and their product with the original parity is $(-1)^{\ell_i}$. In particular their modulus is one. Pairing the endpoints before estimating them exposes the quadratic zero in v.



Pair the two endpoint currents before estimating them.

Figure 1: One coordinate is moved on its whole original interval. Both horizontal arrows give the direction of their integrals. The vertical arrows give the boundary contribution that is subtracted. The two vertical sides are retained and paired by (6.1). Spectator variables remain on their whole real intervals. Only after the distant charges have been summed is a small real cube introduced.

Apply Lemma 3.2 in every spectator coordinate. Using

$$|\sinh(2\pi v)\sinh(\pi n_i N v)| \le C_{\boldsymbol{n}} |n_i| N v^2 e^{\pi |n_i| N v},$$

the absolute value of the weighted paired-side contribution is at most

$$C_{\boldsymbol{n}}|n_i|N\ell_i\, e^{N(\Phi-\gamma_e)}\prod_{j\neq i}|\ell_j|^{-2}\int_0^\tau v^2 e^{-2\pi N(\ell_i-|n_i|/2)v}\,dv. \tag{6.2}$$

For $\ell_i > |n_i|/2+D$, $D \geq 1$, the integral is at most $2/[2\pi N(\ell_i-|n_i|/2)]^3$. The resulting charge sum is finite:

$$\sum_{\ell>|n_i|/2+D}\frac{\ell}{(\ell-|n_i|/2)^3} \leq C\left(D^{-1}+(|n_i|+1)D^{-2}\right). \tag{6.3}$$

For example, write $\ell = (\ell-|n_i|/2)+|n_i|/2$ and compare the sums of the inverse squares and cubes with their decreasing integrals. A change in the integer endpoint only changes the absolute constant. The spectator sums converge by their inverse-square factors.

On the upper horizontal interval use (3.1), with fixed Cauchy neighborhoods for spectator source derivatives. They can be chosen to give total exponential cost at most $2\pi r\tau N$. The shifted Gaussian and Fourier factors in the distinguished coordinate have modulus at most $e^{\pi|n_i|N\tau-2\pi\ell_i N\tau}$. Thus its weighted contribution is at most

$$C_{\boldsymbol{n}}\ell_i\prod_{j\neq i}|\ell_j|^{-2}\exp\left(N\left[\Phi+b_N+2\pi r\tau-2\pi\tau(\ell_i-|n_i|/2)\right]\right). \tag{6.4}$$

Choose the fixed number D so large that $2\pi\tau D > 2\pi r\tau + 2\gamma_e$, and put $M_i = \lceil |n_i|/2+D\rceil$. Summing the geometric tail in (6.4) gives a strict exponential gap. Equations (6.2)–(6.3) give the same for both vertical sides together.

Negative distant charges satisfy the identical estimates, by reflection. Taking a union bound over the distinguished coordinate proves

Proposition 6.1 (The entire distant-charge tail)**.** *For some fixed $\gamma_\infty > 0$ and each fixed sufficiently long twist tuple, the sum of the absolute values of all complete weighted charge terms for which $|\ell_i| > M_i$ for at least one i is at most*

$$C_{\boldsymbol{n}}N^C e^{N(\Phi-\gamma_\infty)}.$$

Every spectator charge is summed over its full nonzero integer range.

The union bound can count a charge more than once; this only enlarges an upper bound. No infinite sum on a sharply truncated small cube has been formed. The absolute convergence already proved and the displayed summable majorants justify each sum and interchange.

7 The remaining charges outside the local range

After Proposition 6.1, only $0 < |\ell_i| \leq M_i$ remain. This is a finite set because the twist tuple was fixed before N varies. Choose a new inner radius b with $0 < b < a/4$, independently of the outer radius B, and let $K_b = [-b,b]^r$.

Suppose one remaining charge satisfies $|\ell_i|/|n_i| > a$. Reflection permits us to take $\ell_i > 0$ for the absolute estimate. On the real domain outside K_b, (1.4) gives a fixed loss. Inside K_b, shift only the ith coordinate by $\mathrm{i}b/|n_i|$. For $|x_j| \leq b$ and $0 \leq v \leq b/|n_i|$, (5.2) gives

$$\operatorname{Re} A_N(x+\mathrm{i}ve_i) \leq A_N(0) - c|x|^2 + Cv^2.$$

The real chirp and charge contribution is

$$-2\pi n_i x_i v - 2\pi\ell_i v \leq -2\pi|n_i|(a-b)v \leq 0. \tag{7.1}$$

On the translated horizontal face, the loss is at least $2\pi b(a-b)$ and the source cost is at most Cb^2/n_0^2. On a vertical side $|x_i| = b$, the fixed loss cb^2 persists, while (7.1) remains nonpositive. Increase n_0 so that these costs are smaller than half the indicated losses. The amplitudes in (1.2) cost only a power of N. Both side faces and the translated interval are therefore bounded by a rate $\Phi - \gamma_{\mathrm{bad}}$ with $\gamma_{\mathrm{bad}} > 0$ fixed.

There are finitely many such bounded charges, so their whole sum has the same strict exponential upper rate, with a constant depending on $\boldsymbol{n}$. Using the inner cube here and the outer cube for the local saddle avoids a conflict between the two contour estimates.

8 The unit charges and proof of the theorem

All remaining charges lie in the local range $|\ell_i|/|n_i| \le a$. They are still a finite set for fixed $\boldsymbol{n}$. Apply Lemma 5.1 inside K_B and (1.4) to its real exterior. Every charge with a magnitude greater than one has, by Lemma 4.2, a positive fixed-tuple gap below the all-unit critical value. The smallest such gap may tend to zero when a twist magnitude tends to infinity; it remains positive in the limit under consideration, where the tuple is fixed.

The 2^r choices of unit signs contribute exactly equally. Indeed, in one coordinate replace ℓ_j by $-\ell_j$ and substitute $\delta_j \mapsto -\delta_j$ on its full symmetric interval. The source and the Gaussian are even, the sine changes sign, and the charge factor changes sign. Those two signs cancel. The parity character is unchanged because the two exponents differ by $2(N+1)\ell_j$. This is an equality of the original integrals, including phase, before any approximation.

For the positive unit vector, the original parity is $(-1)^{(N+1)r}$. Multiplying its local formula by 2^r, the prefactor $\Gamma_N(-\mathrm{i})^r$, and that parity gives (1.8). Choose n_0 large enough that the unit loss $O(\sum_j n_j^{-2})$ is smaller than one quarter of every fixed gap used for the far charges, the bounded nonlocal charges, the side faces and the exterior of K_B. Also require $an_0 \ge 1$. The first charge then lies in the local range and dominates all these parts for sufficiently large N. The other finitely many local charges are controlled by their own strictly positive gaps. Polynomial factors can be absorbed by reducing a positive exponential gap. This proves the error statement in Theorem 1.1.

The limiting coefficient is nonzero by Lemma 5.1, and every factor outside the action has zero exponential rate. Taking absolute values, logarithms and dividing by N proves (1.9). The estimate for the unit action drop follows from Lemma 4.2 after passage to the limit. This completes the proof.

9 Use for a marked knot source

We record the exact interface with the motivating source so that the theorem can be cited without importing a geometric conclusion into an analytic premise. Suppose a marked diagram has r crossing regions and an exact full polynomial source S_N. The marked coefficient and its reduced framing convention are supplied by [1, Theorems 9.1 and 10.1, Corollary 9.2]. Choose the central degree region in every local factor and let P_N be its shared coefficient. This is a linear part of the original source; it need not define an independent invariant under changes of diagram.

In that application the required local expansion has the form

$$C_N(\delta) = \frac{(-1)^{(N-1)r} N^{2r} e^{NA_N(\delta)}}{(2\pi N)^{(r-1)/2}\sqrt{\det H_N(\delta)}}(1 + O(N^{-1})). \tag{9.1}$$

The matrix H_N is the normalized face Hessian on the quotient by common scale. A separate source theorem must construct that quotient, prove its nondegeneracy, choose the positive

complete branch of the determinant root, and establish this expansion on a fixed complex neighborhood. It must also prove the global shared gap (1.4). Nodal values or a formal stationary phase alone would not establish either premise.

If these inputs hold, set

$$\alpha = (3r+1)/2, \qquad \varepsilon_N = (-1)^{(N-1)r}, \qquad a_N = (2\pi)^{-(r-1)/2}(\det H_N)^{-1/2}.$$

The two level-parity factors in (1.8) multiply to one. The complete central filling contribution is therefore

$$\mathcal{I}_N^{\mathrm{central}} = \Gamma_N(-2\mathrm{i})^r\sqrt{2\pi}N^{r+1/2}\frac{\prod_j \sin(2\pi z_{N,j})}{\sqrt{\det H_N(z_N)}\sqrt{\det Q_N}}e^{NS_N} \times (1 + O_{\boldsymbol{n}}(N^{-1}) + O_{\boldsymbol{n}}(e^{-\kappa_{\boldsymbol{n}}N})). \tag{9.2}$$

This derivation includes even and odd levels and their original double-root derivatives. The single reduced mark and the face-lattice normalization belong to the exact finite source theorem and are not inserted again in (9.2).

Two further statements are required for a full knot volume theorem. First, the classical critical equations must construct the smooth filling of the same marked crossing circles and establish

$$2\pi \operatorname{Re} S_0 = \operatorname{Vol}(S^3 \setminus K).$$

Second, the entire difference between the full source and its central part must be bounded after filling. If that remainder has a strictly smaller upper rate, the triangle and reverse triangle inequalities give the full invariant's limit; see [1, Proposition 11.2]. The unit framing character does not change the logarithmic absolute rate. Neither the geometric identity nor that complementary-source estimate is a conclusion of Theorem 1.1.

References

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